

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1134

Barrett 230 kV
March 29, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1134. The Interconnection Customer proposes a 309.7 MW solar photovoltaic facility interconnecting at the Barrett 230 kV substation in the Ohio area. The request was received on January 10, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 414 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,988,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Barrett / Breeze 230.0 kV	Omni-Hackberry / Norwich 500.0 kV	106.2	721	19.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Barrett / Breeze 230.0 kV	Reconductor line	\$5,991,000	33% (\$1,988,000)
	Total allocated to project		\$1,988,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.